

B04 Practical Mastery Check

Closed-book warehouse and quality transfer - non-gating

Non-gating closed-book mastery check

This mastery check never unlocks or blocks Gate A. Save a genuine attempt before opening its protected review.

Scenario - Urban Mobility analytical platform

- Classify booking/payment operations vs historical analytics and write one source-protection boundary.
- Choose a storage/serving pattern for raw ride events, conformed ride facts and governed KPI outputs; state one rejected alternative.
- Design driver history: state business key, warehouse version key, Type 1 vs Type 2 attributes and what makes an effective interval invalid.
- Draw source -> durable landing -> staged/conformed -> serving layers and name the quality gate between each trusted state.
- Choose batch, micro-batch or streaming for three flows with explicit freshness, late-data and replay reasoning.
- Write lineage/ownership metadata for one KPI from source through transform to consumer.
- Define at least six quality rules with severity and action. Include duplicate history, orphan FK, row-count reconciliation and amount reconciliation.
- Write a bounded backfill plan that preserves raw evidence and prevents double publish.

Evidence to save before review

- Architecture/layer sketch.
- History/SCD decision table.
- Quality contract with block/warn/reconcile actions.
- Latency plan and late-data rule.
- Lineage chain with owner.
- Replay/backfill runbook and reconciliation evidence.
- One rejected design with reason.

Pass standard

The check is mastered only when the design is internally consistent, independent controls reconcile, and you can explain which failures block promotion. A polished architecture diagram alone is not a pass.